LOCALIZING A SATURATED MULTIPLICATIVE SYSTEM TO A COVERING TRIANGULATED SUBCATEGORY

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1730-1758, following the statement at 1713-1728.

Editorial note. No omissions: the fully faithful half is displayed as two colimit formulas with the cofinal replacement argued.

Editorial note. Source typo 'is is' kept verbatim.

Allowed standard prerequisites (editorial).

- Localization functor and saturation. proposition-construct-localization (same chapter, cited at ~1732) and Categories, Lemma categories-lemma-what-gets-inverted (cited at ~1734). Identifies S as exactly the arrows inverted by the quotient functor Q.
- Arrows inverted by an exact functor. lemma-triangle-functor-localize (carded L26194, cited at ~1738). Shows S' = arrows of D' inverted by D' -> D -> S^{-1}D is a saturated compatible multiplicative system.
- Universal property and colimit description of morphisms. lemma-universal-property-localization (same chapter, cited at ~1740) and Categories, Remark categories-remark-right-localization-morphisms-colimit (cited at ~1744). Produces the exact functor between localizations and the colimit formula for Hom sets.

Source reference. Original reference 05Q9: focused-seeds/geometry/S9-der-localization-subcategory/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 05Q4: focused-seeds/geometry/S9-der-localization-subcategory/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05R4: sources/stacks-derived/pinned-original-source.tex, source lines 1474–1474. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05R7: sources/stacks-derived/pinned-original-source.tex, source lines 1688–1688. Navigation pointer only: the counted proof

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cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original proposition 05R6: sources/stacks-derived/pinned-original-source.tex, source lines 1577–1577. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{D}$ be a pre-triangulated category and let $\mathcal{D}' \subset \mathcal{D}$ be a full, pre-triangulated subcategory. Let S be a saturated multiplicative system of $\mathcal{D}$ compatible with the triangulated structure. Assume that for each X in $\mathcal{D}$ there exists an $s : X' \to X$ in S such that X' is an object of $\mathcal{D}'$. Then $S' = S \cap Arrows(\mathcal{D}')$ is a saturated multiplicative system compatible with the triangulated structure and the functor*

$$(S')^{-1}\mathcal{D}' \longrightarrow S^{-1}\mathcal{D}$$

is an equivalence of pre-triangulated categories.

Proof. Consider the quotient functor $Q : \mathcal{D} \to S^{-1}\mathcal{D}$ of Proposition 05R6. Since S is saturated we have that a morphism $f : X \to Y$ is in S if and only if $Q(f)$ is invertible, see Categories, Lemma 05Q9. Thus S' is the collection of arrows which are turned into isomorphisms by the composition $\mathcal{D}' \to \mathcal{D} \to S^{-1}\mathcal{D}$. Hence S' is is a saturated multiplicative system compatible with the triangulated structure by Lemma 05R4. By Lemma 05R7 we obtain the exact functor $(S')^{-1}\mathcal{D}' \to S^{-1}\mathcal{D}$ of pre-triangulated categories. By assumption this functor is essentially surjective. Let X', Y' be objects of $\mathcal{D}'$. By Categories, Remark 05Q4 we have

$$\mathrm{Mor}_{S^{-1}\mathcal{D}}(X', Y') = \mathrm{colim}_{s : X \to X' \text{ in } S} \mathrm{Mor}_{\mathcal{D}}(X, Y')$$

Our assumption implies that for any $s : X \to X'$ in S we can find a morphism $s' : X'' \to X$ in S with X'' in $\mathcal{D}'$. Then $s \circ s' : X'' \to X'$ is in S'. Hence the colimit above is equal to

$$\mathrm{colim}_{s'' : X'' \to X' \text{ in } S'} \mathrm{Mor}_{\mathcal{D}'}(X'', Y') = \mathrm{Mor}_{(S')^{-1}\mathcal{D}'}(X', Y')$$

This proves our functor is also fully faithful and the proof is complete. □